Home assignment: day 10 $\to$ day 13

Deadline: 2026-07-15, 16:59. You can use sympy or any other software for symbolic computations.

1. Consider a bracelet with 9 beads. Two bracelets are considered the same if they can be transformed one into another by rotations or flips.
   a) How many rotations (including the trivial 0-degrees rotation) are there?
   b) How many flips (reflections) are there?
   c) Calculate the cycle index for the group of actions.
   d) How many different bracelets can you make with k gemstone types available?
   e) How many different bracelets can you make with exactly 3 diamonds, 3 amethysts and 3 charoits?
2. Watch a Jang Soo Kim video «Polya enumeration theorem».

   https://www.youtube.com/watch?v=1FKAkI-IAkU, .

   We do not cover the proof (12:00-21:00) of the theorem in our short course.

Demo for the quiz on day 11

1. Write the cyclic monomial that corresponds to the transposition $(1)(23)(45)(789)$.
2. You know that the cycle index of the group G that acts on the colourings X is given by

$$CI(z_1, z_2, z_3, z_4) = \frac{1}{24}(z_1^6 + 6z_2z_4 + 8z_3^2 + 9z_1^2z_2^2)$$

   a) How many different colourings in 1, 2, 3 or 4 colours are possible?
   b) How many different colourings in exactly 4 colours are possible?

   You don't need to finalize all arithmetic calculations in the quiz, just write the answer as an expression with numbers.